University to be decided

Dual-Use by Design

Benefits, Harms, and Governance in Decentralized Freedom Technologies

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DOCTORAL THESIS
Draft of 25th October 2026. Replace these pages with initials/initials.pdf.

Abstract

Problem: the gap in knowledge this thesis addresses. Aim and research questions. Data and methods. Main findings of the publications and how they fit together. Contribution to the discipline. At most one page.

Keywords: keyword one, keyword two, keyword three

Tiivistelmä

Tiivistelmä suomeksi. Yksi sivu enintään.

Avainsanat: avainsana yksi, avainsana kaksi

Preface

Acknowledgements to the supervising professor, advisors, pre-examiners, opponent, co-authors, funders, colleagues and others.

Helsinki, 25th October 2026

Akira Taguchi

Contents

Preface vii

List of publications xi

Author's contribution xiii

Abbreviations xv

1 Introduction 1
- 1.1 Background and motivation . . . 1
- 1.2 Research problem and questions . . . 1
- 1.3 Contributions . . . 1
- 1.4 Structure of the thesis . . . 1

2 Theoretical background 3
- 2.1 Key concepts . . . 3
- 2.2 Prior research . . . 3
- 2.3 Analytical framework . . . 3

3 Research design and methods 5
- 3.1 Overall design . . . 5
- 3.2 Data . . . 5
- 3.3 Methods . . . 5
- 3.4 Research ethics and data management . . . 5

4 Summary of the publications 7
- 4.1 Publication I: The Samourai Wallet prosecution . . . 7
- 4.2 Publication II: Relay operators as reluctant moderators . . . 7
- 4.3 Publication III: Building under prosecution risk . . . 7

5 Discussion 9
- 5.1 Integrating the findings . . . 9
- 5.2 Theoretical contributions . . . 9
- 5.3 Implications for policy and practice . . . 9
- 5.4 Limitations . . . 9
- 5.5 Future research . . . 9

6 Conclusion 11

Bibliography 13

Publications 15

Publication I: The Samourai Wallet prosecution 17

Publication II: Relay operators as reluctant moderators 19

Publication III: Building under prosecution risk 21

List of publications

This thesis consists of an overview and of the following publications, which are referred to in the text by their Roman numerals.

I Akira Taguchi. The Samourai Wallet prosecution and its public record. *Planned, 2027.*

II Akira Taguchi. Relay operators as reluctant moderators. *Planned, 2028.*

III Akira Taguchi. Building under prosecution risk. *Planned, 2029.*

Author’s contribution

Publication I I am the sole author of this publication.

Publication II I am the sole author of this publication.

Publication III I am the sole author of this publication.

Abbreviations

EU	European Union
LLM	Large language model
RQ	Research question

1 Introduction

1.1 Background and motivation

The phenomenon, why it matters now, and for whom. Example citations: the potential harms of the Tor anonymity network cluster in free countries (Jardine et al. 2020), Nostr spreads its content over independently run relays (Wei and Tyson 2025), and Bitcoin is an attempt to replace politics with technology (Lehdonvirta 2022, ch. 8).

1.2 Research problem and questions

The gap in existing research, stated as one problem.

RQ1 How do states reach provider-less infrastructure through its developers, and how accurately do public accounts, LLM answers among them, describe that reach? Answered in Publication I.

RQ2 How do the people who run Nostr relays weigh the benefits and harms that pass through them, and what do they do when harmful content arrives? Answered in Publication II.

RQ3 How do developers of non-custodial privacy software read and respond to prosecutions such as those of Samourai Wallet and Tornado Cash? Answered in Publication III.

1.3 Contributions

The thesis's specific scientific contribution, in a few sentences. The pre-examiners and the opponent look for it here.

1.4 Structure of the thesis

Chapter 2 reviews … Chapter 3 describes … Chapter 4 summarises the publications. Chapter 5 integrates the findings, and Chapter 6 concludes.

2 Theoretical background

2.1 Key concepts

Definitions the publications rely on, stated once here.

2.2 Prior research

The literatures the thesis speaks to, and what each leaves open.

2.3 Analytical framework

How the concepts fit together to frame the three publications.

3 Research design and methods

3.1 Overall design

Why this combination of studies answers the research problem.

3.2 Data

Table 3.1: Data and methods by publication.

Publication	Data	Method
I	Court record, agency guidance, LLM answers	Document analysis
II	Interviews ($n = \ldots$), relay policies	Thematic analysis
III	Interviews ($n = \ldots$)	Thematic analysis

3.3 Methods

Analysis procedures, with the detail left to the publications.

3.4 Research ethics and data management

Ethical review, consent, anonymisation, data storage and sharing.

4 Summary of the publications

4.1 Publication I: The Samourai Wallet prosecution

Aim, data and methods, main findings, and the answer to RQ1.

4.2 Publication II: Relay operators as reluctant moderators

Aim, data and methods, main findings, and the answer to RQ2.

4.3 Publication III: Building under prosecution risk

Aim, data and methods, main findings, and the answer to RQ3.

5 Discussion

5.1 Integrating the findings

What the publications show together that none shows alone. This is the argument of the thesis as a whole.

5.2 Theoretical contributions

5.3 Implications for policy and practice

5.4 Limitations

5.5 Future research

6 Conclusion

The answer to the research problem in one or two pages, and the significance of the thesis for the discipline.

Bibliography

Jardine, Eric, Andrew M. Lindner and Gareth Owenson (2020). 'The Potential Harms of the Tor Anonymity Network Cluster Disproportionately in Free Countries'. In: *Proceedings of the National Academy of Sciences* 117.50, pp. 31716–31721. DOI: 10.1073/pnas.2011893117.

Lehdonvirta, Vili (2022). *Cloud Empires. How Digital Platforms Are Overtaking the State and How We Can Regain Control*. Cambridge, MA: The MIT Press. DOI: 10.7551/mitpress/14219.001.0001.

Wei, Yiluo and Gareth Tyson (2025). 'An Empirical Analysis of the Nostr Social Network. Decentralization, Availability, and Replication Overhead'. In: *Proceedings of the ACM on Networking* 3.CoNEXT4, pp. 1–23. DOI: 10.1145/3768994.

Publications

Publication I: The Samourai Wallet prosecution

Akira Taguchi. The Samourai Wallet prosecution and its public record. *Planned, 2027.*

Placeholder. Save the article as publications/I.pdf to include it here.

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Placeholder. Save the article as publications/II.pdf to include it here.

Publication III: Building under prosecution risk

Akira Taguchi. Building under prosecution risk. *Planned, 2029.*

Placeholder. Save the article as publications/III.pdf to include it here.